IGCSE 0607 Practice Questions — E1.16 — Combined Questions & Markscheme

Exponential growth and decay

Total Questions: 21 — **Total Marks:** 114 Marks

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1 A population of bacteria increases exponentially at a constant rate of $p\%$ per hour.
At 09 : 00, the population is 8 000.
At 12 : 00 on the same day, the population is 27 000.

(a) **(a)** Work out the value of p.

$p =$ ..... **[3]**

(b) Work out the population of the bacteria at 14 : 00 on the same day.

..... **[2]**

2 **(a)** The value of a painting increases exponentially by $r\%$ each year.
After 2 years, the total percentage increase in the value of the painting is 44%.
Work out the value of r.

$r =$ ..... **[3]**

(b) The value of a rare stamp increases exponentially.
Its value doubles every 15 years.
The current value of the stamp is \$80.

Work out the value of the stamp after 45 years.

% ..... **[2]**

3 A new car is bought for \$24 000. The value of the car depreciates exponentially at a rate of 8% per year.

Calculate the value of the car after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

4 The population of a city is 125 000.
The population increases exponentially at a rate of 1.8% per year.
Calculate the population of the city after 8 years.
Give your answer correct to the nearest integer.

..... **[3]**

5 A new car is bought for \$26000. The value of the car depreciates exponentially at a rate of 15% each year.

Work out the value of the car after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[3]**

6 A rare stamp is bought for \$450. Its value increases exponentially at a rate of 6.5% per year. Calculate the value of the stamp after 12 years.

\$ ..... **[3]**

7 A machine is bought for \$32 000. It depreciates exponentially at a rate of 12% each year.

Calculate the value of the machine after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

8 The population of a town is 65000.
The population increases exponentially at a rate of 2.8% per year.

(a) Calculate the expected population of the town after 5 years.
Give your answer correct to the nearest integer.

..... **[2]**

(b) Calculate the overall percentage increase in the population over the 5 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

9 The population of a town at the start of 2020 was 42 000.
The population increases at a constant rate of 2.5% each year.

(a) Calculate the population of the town at the start of 2026.
Give your answer correct to the nearest hundred.

..... **[2]**

(b) Calculate the percentage increase in the population from the start of 2020 to the start of 2026.
Give your answer correct to 1 decimal place.

..... % **[2]**

10 A population of bacteria in a dish grows exponentially.

(a) Initially, there are 1500 bacteria in the dish.
The population increases by 8% every hour.
Work out the population of bacteria after 12 hours.
Give your answer correct to the nearest whole number.

..... **[2]**

(b) Work out the least number of complete hours it will take for the initial population of 1500 bacteria to be greater than 3000.

..... hours **[3]**

11 The population of a town was 85 000 on 1 January 2010.
The population increases exponentially at a constant rate of $r\%$ per year.
On 1 January 2022, the population was 112 000.

(a) **(a)** Calculate the value of r, correct to 3 significant figures.

$r =$ ..... **[3]**

(b) Assuming the rate of increase remains the same, calculate the expected population on 1 January 2030.
Give your answer correct to the nearest thousand.

..... **[2]**

12 A car is purchased for \$28 000. Its value depreciates exponentially at a rate of $r\%$ each year. After 4 years, its value is \$16 150.

(a) Calculate the value of r.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

(b) The value of the car continues to depreciate at the same rate.
Calculate the value of the car after a further 3 years.
Give your answer correct to the nearest dollar.

\$ ..... **[2]**

13 A machine costs \$45000. Its value depreciates exponentially at a rate of $x\%$ per year. After 4 years, the value of the machine is \$26000.

(a) Calculate the value of x.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) The machine will be replaced when its value first falls below \$10000.
Find the number of complete years before the machine is replaced.

..... years **[3]**

14 **(a)** The population of a city is 125 000.
It grows exponentially at a rate of 2.4% per year.

Calculate the expected population after 10 years.
Give your answer correct to the nearest hundred.

..... **[3]**

(b) The population of another city was 85 000 in the year 2015.
By 2023, the population had decreased exponentially to 68 000.

Calculate the annual percentage decrease.

..... % **[3]**

15 A delivery van is bought new for \$36 000. Its value depreciates exponentially by 14% each year.

(a) Calculate the value of the van after 4 years.
Give your answer correct to the nearest dollar.

\$ ..... **[2]**

(b) Calculate the number of complete years it takes for the value of the van to first fall below \$10 000.

..... years **[4]**

16 Consider the depreciation of two different vehicles over time.

(a) A car is purchased for \$28 500. Its value depreciates exponentially at a rate of 12% per year.
Calculate the value of the car after 5 years, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

(b) A van, initially purchased for \$35 000, depreciates exponentially to a value of \$15 200 after 4 years.
Calculate the annual percentage rate of depreciation for this van.

..... % **[4]**

17 The mass, M grams, of a radioactive substance decays exponentially at a rate of $r\%$ per day.
After 2 days, the mass is 80 grams.
After 5 days, the mass is 40.96 grams.

(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) Find the initial mass of the substance.

..... grams **[2]**

(c) Calculate the number of complete days from the start when the mass is first less than 10 grams.

..... days **[3]**

18 A radioactive substance decays exponentially.

(a) The mass of the substance decreases by 12% every day.
The initial mass is 500 grams.

Calculate the mass remaining after 1 week (7 days).

..... g **[2]**

(b) After d complete days, the mass of the substance falls below 50 grams.

Find the smallest integer value of d.

$d =$ ..... **[3]**

(c) A different radioactive substance decays exponentially such that its mass halves every 15 days.

Calculate the percentage decrease in its mass each day.

..... % **[3]**

19 The mass, M grams, of a radioactive substance remaining after t hours is modelled by the formula

$$M = M_0 \times k^{-t}$$

where M_0 and k are positive constants.
At $t = 2$, the mass is 256 grams.
At $t = 5$, the mass is 108 grams.

(a) Show that $k = \frac{4}{3}$.

[3]

(b) Find the value of M_0.

$M_0 =$ ..... **[2]**

(c) Calculate the value of t when the mass of the substance has decreased to 10% of its initial mass M_0.
Give your answer correct to 2 decimal places.

$t =$ ..... **[3]**

20 Alex and Ben each invest \$4000 for 10 years. Alex invests his \$4000 in an account paying compound interest at a rate of 4.5% per year.
Ben invests his \$4000 in an account paying compound interest at a rate of r% per year.

(a) Calculate the total amount in Alex's account after 10 years.

\$ ..... **[2]**

(b) After 10 years, Ben has \$500 more than Alex.
Calculate the value of r.

$r =$ ..... **[4]**

(c) A different investment of \$2500 grows exponentially. It doubles in value every 8 years.
Calculate its value after 20 years.

\$ ..... **[2]**

21 The population of a species of insect increases exponentially.
Initially, there are 2500 insects.
After 6 weeks, there are 9100 insects.

(a) Calculate the percentage increase in the population per week.

..... % **[4]**

(b) Calculate the insect population after 10 weeks, giving your answer correct to the nearest 100.

..... **[2]**

(c) Find the number of complete weeks it takes for the initial population to increase to over 50 000.

..... weeks **[3]**

1 **(a)** **(a)** $8000(1+\frac{p}{100})^3 = 27000$ **[M1]**

$1+\frac{p}{100} = \sqrt[3]{\frac{27000}{8000}} = \frac{3}{2}$ **[M1]**

50 **[A1]**

(b) $27000 \times (1.5)^2$ or $8000 \times (1.5)^5$ **[M1]**

60 750 **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** $\left(1+\frac{r}{100}\right)^2 = 1.44$ or $x^2 = 1.44$ **[M1]**

$1+\frac{r}{100} = 1.2$ or $x = 1.2$ **[M1]**

20 **[A1]**

(b) Number of doubling periods $= 45 \div 15 = 3$ or 80×2^3 **[M1]**

640 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 M1 for $24000 \times (1-0.08)^5$ or 24000×0.92^5

A1 for 15817.96...

15818 **[B1]**

[Back to Question] *[Total Marks: 3]*

4 $125\,000 \times (1+0.018)^8$ or $125\,000 \times 1.018^8$ [M2]
(If not M2, award M1 for 1.018^8 or $125\,000 \times 1.018^k$ where $k > 1$)

144 183.6... **[A1]**

144 184 **[B1]**

[Back to Question] *[Total Marks: 3]*

5 $26000 \times (1-0.15)^4$ or 26000×0.85^4 **[M2]**

If M0, allow [M1] for 26000×0.85^k where $k > 1$

13572 **[A1]**

[Back to Question] *[Total Marks: 3]*

6 $450 \times \left(1+\frac{6.5}{100}\right)^{12}$ **[M1]**

450×1.065^{12} **[M1]**

958.01 (accept 958) **[A1]**

[Back to Question] *[Total Marks: 3]*

7 $32000 \times (1 - 0.12)^5$ oe **[M2]**

For 32000×0.88^n where $n > 1$, award **[M1]**

16887 **[A1]**

[Back to Question] *[Total Marks: 3]*

8 **(a)** $65000 \times \left(1 + \frac{2.8}{100}\right)^5$ or 65000×1.028^5 **[M1]**

74624 **[A1]**

(b) $\frac{74624 - 65000}{65000} \times 100$ or $(1.028^5 - 1) \times 100$ **[M1]**

14.8 **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** $42\,000 \times 1.025^6$ $(= 48\,707.12...)$ **[M1]**

48 700 **[A1]**

(b) $(1.025^6 - 1) \times 100$ or $\frac{48\,707.12... - 42\,000}{42\,000} \times 100$ **[M1]**

16.0 (15.969...) **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** 1500×1.08^{12} **[M1]**

3777 **[A1]**

(b) $1500 \times 1.08^n > 3000$ or $1.08^n = 2$ **[M1]**

Trial with values (e.g., $1.08^9 = 1.999...$ and $1.08^{10} = 2.158...$) or uses logarithms (e.g., $n = \frac{\log(2)}{\log(1.08)} = 9.006...$) **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** **(a)** $85000(1 + \frac{r}{100})^{12} = 112000$ **[M1]**

$1 + \frac{r}{100} = \left(\frac{112}{85}\right)^{\frac{1}{12}}$ **[M1]**

2.33 (from 2.327...) **[A1]**

(b) $85000 \times \left(1 + \frac{2.327...}{100}\right)^{20}$ or $112000 \times \left(1 + \frac{2.327...}{100}\right)^{8}$ **[M1]**

135 000 **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** $28\,000 \times \left(1 - \frac{r}{100}\right)^4 = 16\,150$ **[M1]**

$\left(1 - \frac{r}{100}\right) = \sqrt[4]{\frac{16\,150}{28\,000}}$ or $\sqrt[4]{0.57678...}$ **[M1]**

12.9 (or 12.88 to 12.89) **[A1]**

(b) $16\,150 \times (1 - 0.1289...)^3$ or $28\,000 \times (1 - 0.1289...)^7$ **[M1]**

10 677 **[A1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** $45000 \times \left(1 - \frac{x}{100}\right)^4 = 26000$ **[M1]**

$1 - \frac{x}{100} = \sqrt[4]{\frac{26000}{45000}}$ or $1 - \frac{x}{100} = 0.8711...$ **[M1]**

12.9 **[A1]**

(b) $45000 \times (0.8711...)^n < 10000$ or $(0.8711...)^n < \frac{2}{9}$ **[M1]**

Testing values or using logarithms: $n = 10 \implies \$11342$ and $n = 11 \implies \$9881$ **[M1]**

11 **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** M1 for $125000 \times \left(1 + \frac{2.4}{100}\right)^{10}$ or 125000×1.024^{10}

A1 for 158461.5 . . .

158500 **[B1]**

(b) M1 for $85000 \times \left(1 - \frac{r}{100}\right)^8 = 68000$

M1 for $1 - \frac{r}{100} = \sqrt[8]{\frac{68000}{85000}}$ or $0.8^{\frac{1}{8}}$

2.75 (accept 2.750 to 2.751) **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** $36\,000 \times 0.86^4$ $(= 19\,692.29...)$ **[M1]**

19 692 **[A1]**

(b) $36\,000 \times 0.86^n < 10\,000$ or $0.86^n = \frac{10\,000}{36\,000}$ **[M1]**

Applies logarithms or systematic trials: $n = \frac{\log(10/36)}{\log(0.86)}$ or $0.86^8 = 0.2992...$,

$0.86^9 = 0.2573...$ **[M1]**

$n = 8.49...$ **[A1]**

9 **[A1]**

[Back to Question] *[Total Marks: 6]*

16 **(a)** $28\,500 \times \left(1 - \frac{12}{100}\right)^5$ **[M1]**

$28\,500 \times 0.88^5$ **[M1]**

15 040 **[A1]**

(b) $35\,000 \times \left(1 - \frac{r}{100}\right)^4 = 15\,200$ **[M1]**

$\left(1 - \frac{r}{100}\right)^4 = \frac{152}{350}$ **[M1]**

$1 - \frac{r}{100} = \sqrt[4]{\frac{152}{350}} = 0.8123\ldots$ **[M1]**

18.8 (accept 18.76 to 18.77) **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** Let $k = 1 - \frac{r}{100}$. Then $M_0k^2 = 80$ and $M_0k^5 = 40.96$ **[M1]**

$k^3 = \frac{40.96}{80} = 0.512 \implies k = \sqrt[3]{0.512} = 0.8$ **[M1]**

$1 - \frac{r}{100} = 0.8 \implies r = 20$ **[A1]**

(b) $M_0 \times 0.8^2 = 80$ or $M_0 \times 0.64 = 80$ **[M1]**

125 **[A1]**

(c) $125 \times 0.8^n < 10$ or $0.8^n < 0.08$ **[M1]**

Testing values: $n = 11 \implies 0.8^{11} = 0.0858...$ and $n = 12 \implies 0.8^{12} = 0.0687...$ **[M1]**

12 **[A1]**

[Back to Question] *[Total Marks: 8]*

18 **(a)** M1 for $500 \times (1 - 0.12)^7$ or 500×0.88^7

204.3 (accept 204 or $204.33\ldots$) **[A1]**

(b) M1 for $500 \times 0.88^d < 50$ or $0.88^d = 0.1$

M1 for evaluating trials (e.g. $0.88^{18} = 0.1001\ldots$ and $0.88^{19} = 0.0881\ldots$) OR using logarithms $d > \frac{\log(0.1)}{\log(0.88)}$

19 **[A1]**

(c) M1 for $\left(1 - \frac{r}{100}\right)^{15} = 0.5$

M1 for $\sqrt[15]{0.5}$ or $0.5^{\frac{1}{15}}$

4.52 (accept 4.515 to 4.516) **[A1]**

[Back to Question] *[Total Marks: 8]*

19 **(a)** $M_0k^{-2} = 256$ and $M_0k^{-5} = 108$ **[M1]**

Divides equations: $\dfrac{M_0k^{-2}}{M_0k^{-5}} = \dfrac{256}{108} \implies k^3 = \dfrac{64}{27}$ **[M1]**

$k = \sqrt[3]{\dfrac{64}{27}} = \dfrac{4}{3}$ **[A1]**

(b) $M_0 \times \left(\dfrac{4}{3}\right)^{-2} = 256$ or $M_0 = 256 \times \dfrac{16}{9}$ **[M1]**

455.1 (455.111... or $\dfrac{4096}{9}$) **[A1]**

(c) $0.1M_0 = M_0\left(\dfrac{4}{3}\right)^{-t}$ or $\left(\dfrac{3}{4}\right)^{t} = 0.1$ **[M1]**

$t = \dfrac{\log(0.1)}{\log(0.75)}$ or $t = \dfrac{\log(10)}{\log(4/3)}$ **[M1]**

8.00 (8.0039...) **[A1]**

[Back to Question] *[Total Marks: 8]*

20 **(a)** $4000 \times \left(1 + \dfrac{4.5}{100}\right)^{10}$ or 4000×1.045^{10} **[M1]**

6211.88 (or 6211.87...) **[A1]**

(b) $6211.88 + 500$ or 6711.88 **[B1]**

$4000 \times \left(1 + \dfrac{r}{100}\right)^{10} = 6711.88...$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[10]{\dfrac{6711.88...}{4000}}$ **[M1]**

5.31 (or 5.31...) **[A1]**

(c) $2500 \times 2^{\frac{20}{8}}$ or $2500 \times 2^{2.5}$ **[M1]**

14 142 (or 14 142.13...) **[A1]**

[Back to Question] *[Total Marks: 8]*

21 **(a)** $2500 \times \left(1 + \dfrac{r}{100}\right)^{6} = 9100$ **[M1]**

$\left(1 + \dfrac{r}{100}\right)^{6} = 3.64$ **[M1]**

$1 + \dfrac{r}{100} = 3.64^{\frac{1}{6}} = 1.2403\ldots$ **[M1]**

24.0 (accept 24.03 . . .) **[A1]**

(b) 2500×1.2403^{10} or 9100×1.2403^{4} **[M1]**

$21\,500$ **[A1]**

(c) $2500 \times 1.2403^{n} > 50\,000 \implies 1.2403^{n} > 20$ **[M1]**

$n \times \log(1.2403) > \log(20) \implies n > 13.91\ldots$ (or equivalent trial and error) **[M1]**

14 **[A1]**

[Back to Question] *[Total Marks: 9]*